

RedRock Email Classifier

Classifies inbound RedRock client emails into one of five business categories (Account Management, Investment Advisory, Loan Processing, Insurance Claims, Other) so they can be routed to the correct department. Outputs a predicted category and a calibrated confidence score per email.

See [UoW.md](#) for the full unit-of-work write-up: approach comparison, evaluation results, statistical rigor (bootstrap confidence intervals, calibration, risk-coverage, conformal-prediction feasibility), a comparative review of an alternative approach, and regulatory considerations.

Setup

Requires Python 3.10 or later.

```
python -m venv .venv
source .venv/bin/activate    # Windows: .venv\Scripts\activate
pip install -r requirements.txt
```

The embedding classifier downloads a small (roughly 90MB) sentence encoder model (`sentence-transformers/all-MiniLM-L6-v2`) from Hugging Face on first run; this requires internet access once, then works offline.

Running it

Everything (data loading, cross-validation, calibration, and final test-set predictions) runs end to end with one command:

```
python src/evaluate.py
```

This will: 1. Load the 44 labeled training emails and the 12 unlabeled test emails from `data/`. 2. Run 5-fold stratified cross-validation, repeated 10 times, for both classifiers on the training set, and a nested calibration check to confirm the calibration gain is not an artifact of leakage. 3. Fit each classifier on all 44 labeled emails, calibrate its confidence

score, and predict the 12 test emails. 4. Write, into `results/` :-

`predictions_baseline_tfidf_logreg.csv` - `predictions_embedding_centroid.csv` , this is the primary submission - `predictions.csv` , a copy of the primary submission's predictions - `evaluation_results.json` (full metrics: per-class breakdown, calibration before and after temperature scaling, bootstrap confidence intervals, risk-coverage table, conformal-prediction feasibility, and cost-weighted error) - `comparison.csv` (headline comparison table) - `evaluation_report.txt` (the same results in a human-readable form)

Each predictions CSV has the columns required by the spec: `email_id` , `predicted_category` , `confidence_score` . The confidence score is the temperature-calibrated probability, not the raw model output.

Running the tests

```
python -m pytest tests/ -v
```

Project layout

```
src/  
  ingestion.py  
    parses the HTML emails into structured records  
  classifier_baseline.py  
    TF-IDF (word and character n-gram) + Logistic Regression  
  classifier_embedding.py  
    sentence-embedding nearest-centroid classifier (primary)  
  calibration.py  
    temperature scaling, calibration metrics, bootstrap CI,  
    risk-coverage, conformal feasibility, cost-weighted error  
  evaluate.py  
    cross-validation + calibration + final predictions, single entry point  
data/  
  train/, test/, train_labels.csv  
    provided sample data  
tests/  
  pytest suite for ingestion, both classifiers, calibration, and the eval  
  harness
```

```
results/  
generated by evaluate.py (not committed pre-run)
```